

The Distorted Time Framework

Volume II — *The Shape of Time*

Gravity, inertia, and the resolution of the black-hole singularity

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Abstract

With time’s local rate u as the single primitive (Volume I), gravity is the shape of that rate: a slow body falls down its gradient, and the field law $\nabla^2 u = 4\pi G\rho/c^2$ is the rate’s, not a graviton’s. The reconstructed metric matches general relativity in every tested regime (the Isenberg–Wilson–Mathews conformally-flat scheme, exact to 1PN), and the frame field radiates exactly two tensor polarisations — so the tested content is GR’s, and by Deser’s uniqueness the two are indistinguishable wherever the spin-2 field is defined. The volume’s distinctive claim lies where that field is *not* defined: at Planck curvature the emergent description of time **de-emerges**, replacing the singularity with a finite core and a stable Planck-mass remnant — a *boundary conjecture* whose signatures (no echoes, no bounce) differ sharply from the fuzzball and the Planck star. The rotating exterior is matched to Kerr (forward through $O(a^3)$, all orders inherited via Deser). Stated plainly: the singularity results sit at the epistemic level of a boundary conjecture, not a derived dynamics.

Keywords: black-hole singularity; time de-emergence; general relativity; Kerr metric; gravitational-wave polarisations; conformal flatness.

The three commitments (from the Core, Volume I). This volume builds on three, argued there and used here without re-argument (full recap: Appendix A):

- **1. Time is primitive** — its local rate $u(\mathbf{x})$ is the one physical thing: Row 0 (timeless substrate, $\hat{H}\Psi = 0$, energy conserved as a constraint) $\rightarrow$ Row 1 (the clock, where gravity and inertia live) $\rightarrow$ Row 2 (space).
- **2. Space is time’s radiation** — $\lambda_C = cT_C$; time takes $g_{00} = -u^2$ and space the complement $\gamma_{ij} = (2 - u^2)\delta_{ij}$, while u itself does *not* propagate ($\pi_u \equiv 0$). Hence zero new degrees of freedom, exact Lorentz ($\alpha_1 = \alpha_2 = 0$), and a radiative content of exactly two — read forward from what space is made of (Appendix A). These carry the polarisation count and the singularity alike.
- **3. There is a proper now** — the slice that solves Row-0’s energy constraint (constant- u /CMC), real but empirically silent.

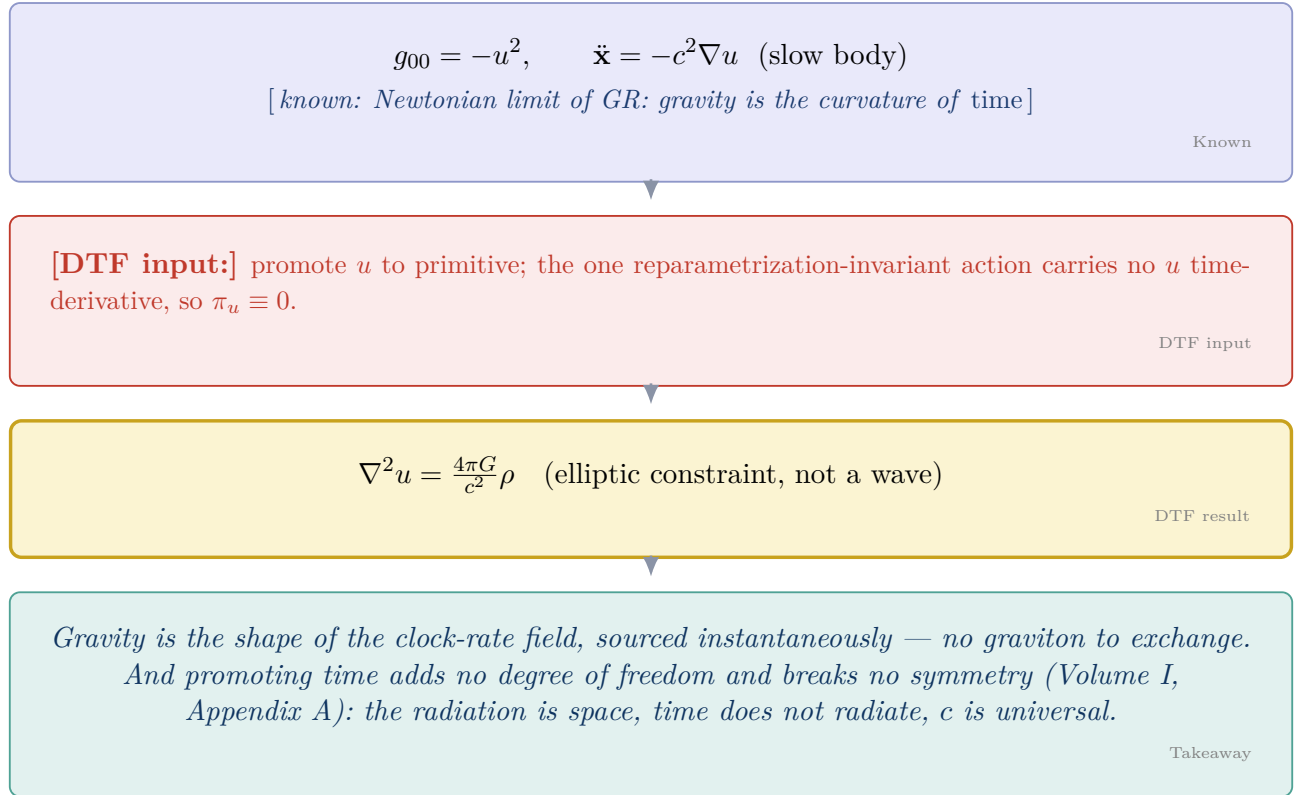
Here the rate is *sourced* by matter (gravity) and pushed to where it *de-emerges* (the singularity). What stays contingent is only the conformally-flat scheme this volume uses to compute the metric beyond 1PN — a limit on calculation, not on the commitments.

General relativity is our most successful theory of gravity, and it predicts its own failure: at the centre of a black hole the curvature diverges and the theory stops. This volume says *what that failure is*. With time’s rate as the primitive, the singularity is not a place where curvature becomes infinite — it is where the emergent description of time **de-emerges**, at Planck curvature, and “what is at $r = 0$ ” is a question asked in a language that has ceased to exist there. What replaces the singularity is a

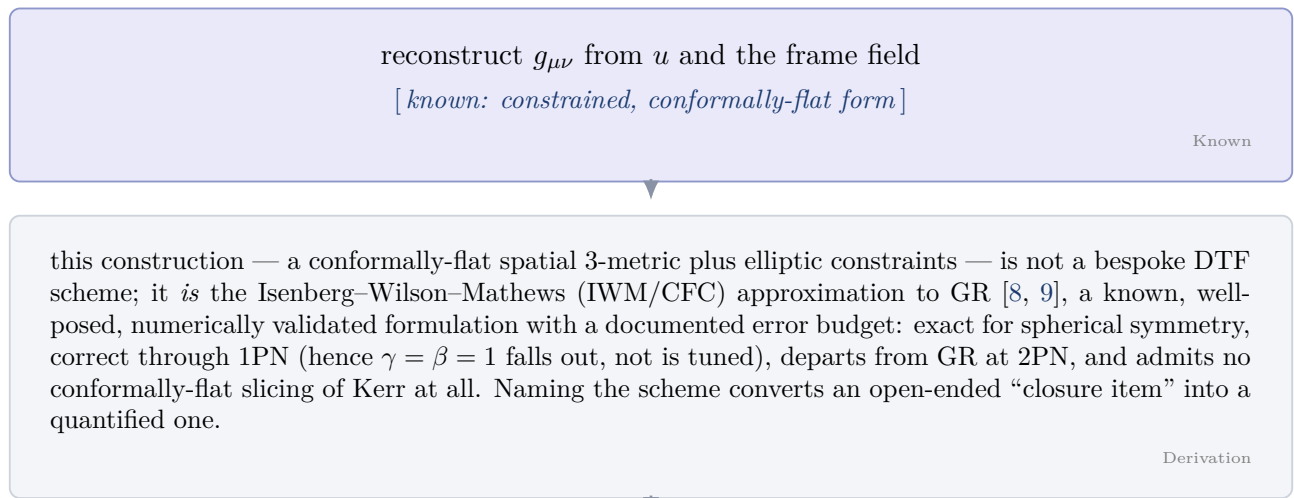
finite core, and it makes predictions that differ, sharply, from every other proposal. First, though, the gravity the singularity lives in.

Chain 1 *Gravity is the shape of the clock field*

A slow body falls down the gradient of time’s rate. Make that rate primitive, and gravity is its shape — with no force carrier.

**Chain 2** *General relativity, matched in the tested regime*

The reconstructed metric is GR’s in every tested regime — by construction, not by tuning; beyond it, matched only where the scheme is known to hold.



$$\gamma = \beta = 1, \quad c_{\text{GW}} = c \quad (\text{IWM/CFC, exact to 1PN})$$

DTF result

The PPN parameters [3] and the GW speed come out as GR's at the order IWM/CFC is known to reproduce it. Honest caveat: the spatial metric is reconstructed in this conformally-flat approximation with a documented g_{rr} deviation beyond 1PN (the closure item, sharpened above) — so this is GR proposed as what the clock field is made of at the tested level, not a claim of exact recovery. The binary-pulsar orbital decay — matched to $\sim 0.2\%$ against PSR B1913+16 [2] — is a radiation-reaction effect, which the waveless conformally-flat scheme does not itself produce: it is GR's quadrupole result, and DTF inherits it not through the CFC reconstruction but through the Deser closure (box after Chain 8) — the radiative sector is a two-polarisation spin-2 field, so its matter coupling, and hence the $G/5c^5$ quadrupole emission, are GR's (a mode count alone would not fix the prefactor). A genuinely independent DTF quadrupole computation from the frame field's own dynamics is not written out in this trilogy.

Takeaway

Chain 3 *Space radiates — two tensor polarisations*

The radiative sector is space. How many ways it can wave follows from what space is made of — not from a projector chosen in advance.

$$h_{ij} \text{ symmetric : } \frac{d(d+1)}{2} = 6 \text{ components at } d = 3$$

[known: the extent structure — what assigns a length to each direction]

Known

space is compiled from extents (Volume I, Chain 5), so the object that can modulate is h_{ij} , and the frame's three internal rotations are redundancy in it (they leave $g_{ij} = e_i^a e_j^a$ fixed at all orders). Split the six by propagation direction — 2 scalar, 2 vector, 2 tensor — and the framework removes four of them: the scalars because a uniform rescaling of extents *is* a clock-rate change, which the radiation rule forbids from propagating (Volume I, Chain 5) and the gravity field law fixes elliptically (Chain 1); the vectors because space is compiled, so relabelling points within a slice is not a physical change.

Derivation

$$6 - 2 (\text{scalar}) - 2 (\text{vector}) = 2 \text{ transverse-traceless}$$

`DTF_radiative_content_constructive.py` → 2, forward from the premises

DTF result — verified

exactly two tensor polarisations, no breathing mode

DTF result

This is the framework's cleanest falsifiable stake: a detected scalar/breathing gravitational-wave polarisation would refute it. Time (the scalar u) does not radiate — and the count is now read forward from what space is, with the familiar TT projector appearing as a consequence of the split rather than as its premise (Volume I, §A.4).

Takeaway

One object, seen here as: inertia & gravity

One rate ω_C : it *is* the inertia (linear in m , definitional), and it *sources* the gravitational well in u (the G -coupled $\sim (m/m_P)^2$ dent), whose extremum is free fall. The equivalence principle is that single identity — read forward into the chain below, where inertia is the rate, not a wake.

Chain 4 *Inertia — the rate, not a wake*

Inertia needs no mechanism. Mass is the clock's ticking rate; inertia is that rate, carried by a moving body.

$$\mathbf{p} = \nabla S = \gamma m \mathbf{v}, \quad E = -\partial_t S = \gamma m c^2$$

[known: Hamilton–Jacobi: the moving clock's momentum and energy]

Known

inertia = the rate ω_C itself ($m = \hbar\omega_C/c^2$), linear in m , no G

DTF result

There is no self-energy to compute. Mass is the ticking rate (Volume I, “a note on mass”); inertia is that rate carried by a moving body — linear in m , $\hbar$ -set, carrying no G . The coefficient mc^2 is not a debt but a unit conversion ($E = \hbar\omega_C$). A self-field reaction to the body's own u -well could not be the source: that well is G -coupled and $\sim (m/m_P)^2$ shallow — some forty-five orders too weak to be inertia, which in any case needs no such mechanism.

What remains owed is only the pattern of ticking rates (the mass hierarchy) — a free parameter for every theory alike; deferred, not claimed.

Takeaway

Chain 5 *The singularity: time de-emerges*

Curvature does not diverge — the description does. Where the clock can no longer complete a tick, Row-1 time ends.

$$K = R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} = \frac{48 (GM/c^2)^2}{r^6}, \quad L_{\text{curv}} = K^{-1/4}$$

[known: Kretschmann invariant, Schwarzschild]

Known

[DTF input:] Row-1 time de-emerges where the clock loses definiteness: one coherent tick no longer fits in a curvature time, $N \equiv L_{\text{curv}}/\ell_P \rightarrow 1$ (Planck curvature, coefficient-free). The Planck-curvature *threshold* is imported — most quantum-gravity programmes agree curvature saturates there; what is *native* is the interpretation: de-emergence of Row-1 time, not a bounce or a new phase.

DTF input

$$r_* = 48^{1/6} \ell_P (M/m_P)^{1/3}, \quad \frac{r_*}{r_s} \simeq 0.95 (M/m_P)^{-2/3}$$

`DTF_singularity_deemergence.py` → deep interior, fat core

DTF result — verified

a finite core at r_* , never the point $r = 0$

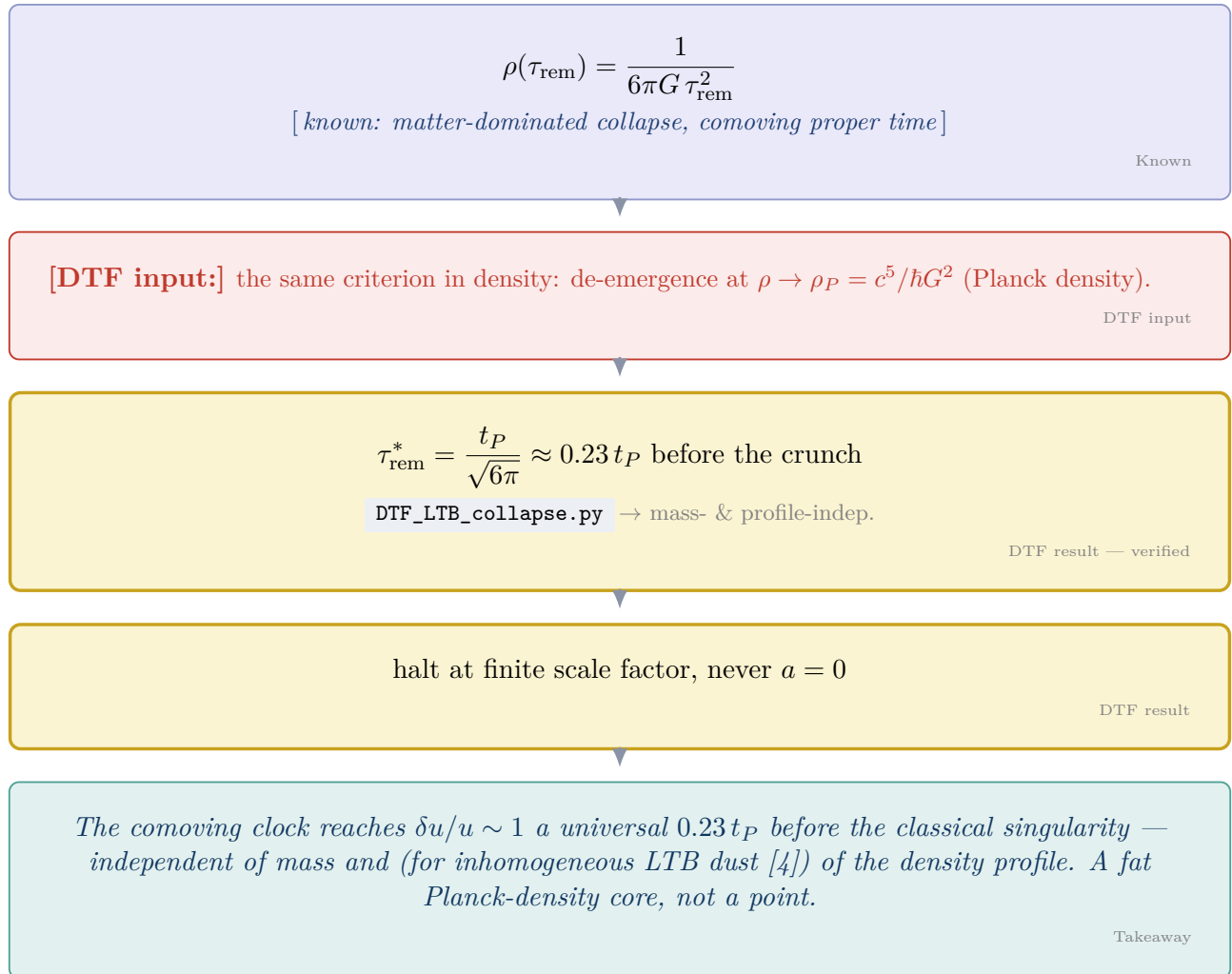
DTF result

The de-emergence surface sits parametrically inside the horizon ($\sim 10^{-26} r_s$ for a solar mass); the horizon stays curvature-regular. The singularity is a question in a language that has ceased to exist — and what is left at the core is Row 0 showing through: the timeless substrate, uncovered where Row-1 time ends. The ontology closes on itself.

Takeaway

Chain 6 *The interior: a clock that halts before the crunch*

Read through a geodesic comoving clock, collapse stops a Planck time short of the classical singularity.



Chain 7 *Observables — how the finite core differs from its rivals*

A finite core in place of a singularity has consequences at both ends of a black hole's life.

$$r_s \propto M, \quad R_* = (M/\rho_P)^{1/3} \propto M^{1/3}$$

[known: horizon and core shrink at different rates]

Known

$$M_{\text{rem}} = \frac{m_P}{\sqrt{8}} \approx 0.35 m_P, \quad \text{no echoes, no bounce}$$

`DTF_singularity_observable.py` → stable Planck-mass relic

DTF result — verified

a stable, horizonless Planck-mass remnant

DTF result

Evaporation halts when the core meets the shrinking horizon — no complete evaporation, no final burst. The structure sits at $\sim 10^{-26} r_s$, so no post-ringdown echoes; time de-emerges rather than bouncing, so no radio/gamma bounce transient. Whether a population of such relics could contribute to dark matter is a separate question this volume does not assess.

Takeaway

Signature	DTF (de-emergence)	Rivals
Post-ringdown GW echoes	none (regular horizon)	fuzzball/firewall[5, 6]: yes
Evaporation endpoint	Planck remnant $\sim 0.35 m_P$	complete / burst
Primordial-hole fate	stable Planck-mass relic	Planck star[7]: bounces
Radio/gamma bounce transient	none	Planck star: yes

Chain 8 *The rotating sector — Kerr matched (exterior; forward through $O(a^3)$, higher orders inherited)*

Rotation is the one place the conformally-flat reconstruction stops. Sourced forward from DTF's own sectors — not read off Kerr — it is reached even so: frame-dragging is the proper now dragged, and the spatial geometry follows to second order in spin.

$$\text{tr } K = -\frac{1}{N} D_i \beta^i = 0$$

[known: a stationary axisymmetric const- u slice is maximal — a legitimate proper now]

Known

[DTF input:] rotation enters only through the shift β^i , sourced by the matter momentum ρv^i *alone* ($\pi_u \equiv 0$: no clock current to be dragged alongside it).

DTF input

$$\omega = \frac{2GJ}{c^2 r^3} \text{ (DTF, forward)} = \omega_{\text{Kerr}}^{\text{far}}, \quad N_{\text{dof}} = 2$$

`DTF_kerr_constructive.py` → frame-dragging is native; rotation adds no mode

DTF result — verified

$$C_{ijk} = 0 \iff \text{conformally flat} \iff \text{no physical tensor content}$$

[known: the coordinate-invariant obstruction (Cotton tensor)]

Known

compute C_{ijk} of the Kerr spatial 3-geometry on that slice, order by order in spin: it is zero at $O(a^0)$ (Schwarzschild) and $O(a^1)$ (frame-dragging does not curve space), and *first nonzero at $O(a^2)$* . So the tensor sector must switch on exactly there — and its source is the *quadratic* of the $O(a)$ shift, a gravitomagnetic dipole (hence angular content $\ell = 0, 2$).

Derivation

$$\delta R[g_2] = -\delta^2 R[g_1] \quad \text{balances to } 10^{-126}$$

`DTF_kerr_oa2_sufficiency.py` → frame-drag² sources the $O(a^2)$ TT mode

DTF result — verified

DTF = Kerr through $O(a^2)$, coefficient-exact

DTF result

The frame-drag² source, pushed through DTF's radiative tensor operator, reproduces Kerr's second-order geometry exactly; by uniqueness of the elliptic problem the forward solve is Kerr. This is proven explicitly here at $O(a^2)$; the extension to all orders is inherited from the Deser bootstrap (box below), not computed order by order. It is the exterior rotating geometry (frame-dragging, ergosphere $|\beta| = uc$); rotating collapse is a separate question, not claimed here.

Takeaway

What the rotating sector rests on — and why it closes to all orders.

- **The reframe.** Conformal flatness is the *computational* (IWM/CFC) scheme, not the ontology. Kerr admits a maximal ($K = 0$) slice — the very slice DTF calls the proper now — and Garat–Price [11] forbids only a *conformally-flat* one. The theorem bounds the old reconstruction, not the framework.
- **The operator is not a premise — it is Lorentz invariance.** That DTF's *stationary* tensor operator equals its *radiative* one is not an extra assumption: universal c (Volume I, Chain 5) makes the wave operator $\square = -c^{-2}\partial_t^2 + \nabla^2$ the *only* second-order operator the transverse-traceless frame field can obey. The linearised field operator on a TT mode is exactly $-\frac{1}{2}\square h_{ij}$ (two propagating polarisations), and its stationary reduction ($\partial_t \rightarrow 0$) is $-\frac{1}{2}\nabla^2$ — the very operator that balanced the frame-drag² source. Radiative and near-zone are one operator, evaluated at two values of ∂_t . `DTF_kerr_premise_oneoperator.py` $\rightarrow$ $\delta R[h^{\text{TT}}] = -\frac{1}{2}\square h$, $\text{stationary} = \nabla^2$
- **Native through $O(a^3)$; all orders by Deser — and the two are graded.** The operator ($\square$, all orders) and the quadratic vertex (the radiative milestone) fix the solution *through* $O(a^2)$; $O(a^3)$ is the first order whose source is *cubic* (frame-drag sector, by parity), probing the cubic vertex. That vertex is not free: DTF's frame field is established to be a two-polarisation spin-2 metric (App. A / Volume I) whose internal $SO(3)$ is gauge (invisible in $g_{ij} = e_i^a e_j^a$ to all orders) and whose clock u is non-propagating ($\pi_u \equiv 0$), so its physical propagating content is precisely a consistent, Lorentz-invariant, two-derivative, two-polarisation spin-2 field — for which Deser's self-coupling bootstrap [10] makes general relativity the *unique* completion, every vertex GR's. **The grading matters, and we state it:** the $O(a^2)$ and $O(a^3)$ agreements are *computed forward* in DTF (closing to 10^{-126} and 10^{-131}); the extension to *all* orders is *inherited* from Deser's theorem — which any two-polarisation spin-2 theory shares — not an order-by-order DTF calculation. Granted that inheritance, DTF = Kerr to all orders in the exterior, and the rotating sector adds no distinctive signature (the distinctive stakes remain the GW polarisations and the black-hole cores). The one honest residual is the rotating *interior* (rotating collapse / de-emergence), untouched here. `DTF_kerr_deser_closure.py` $\rightarrow$ $\text{native} \leq O(a^3)$; all orders inherited via Deser (2-pol spin-2)

What Deser closes, and what it does not. The bootstrap [10] is a theorem about a *field*: any consistent, Lorentz-invariant, two-derivative, two-polarisation spin-2 field completes uniquely to general relativity. DTF's frame field is such a field, so wherever that field is defined DTF's vertices are GR's — which is what closes the rotating exterior to all orders, and *equally* what makes the gravitational sector observationally indistinguishable from GR in every tested regime.

It follows that no result of this volume's core sector can be a modification of GR's field equations, and none is offered as one. De-emergence is a claim about *where the spin-2 description ceases to be defined* — at $\delta u/u \sim 1$ — and Deser's theorem, like every perturbative uniqueness result, is silent there. This places the finite core, the halt, and the remnant at the epistemic level of a *boundary conjecture*: the same level as the Planck star and the fuzzball, differing in *what* is claimed to lie at the boundary, not in derivational status. The comparison table (Chain 7) is a comparison of boundary conjectures by signature, not a derivation discriminating against rivals.

What it stakes, and what it owes.

- **Falsifiable stakes, and which discriminate.** Two tensor GW polarisations, no breathing mode — a falsifiable stake, but one *shared with* GR, not a discriminator. The echo-free finite core, the stable Planck-mass remnant, and the absence of a bounce distinguish DTF's boundary conjecture from rival *conjectures* (fuzzball, Planck star) *by signature* — a comparison of conjectures at one epistemic level, not a derivation ranking DTF above them (see the Deser box after Chain 8).
- **Honest limits.** What is removed is the *curvature* singularity and the classical endpoint; whether the object is geodesically *complete* — as opposed to a worldline exiting Row 1 — is a separate, open question. Every numerical coefficient (r_* , the $0.23 t_P$ halt, M_{rem}) is an $O(1)$ illustration in a tractable model; the load-bearing claim (de-emergence at Planck curvature) is coordinate-invariant and coefficient-free.
- **Rotation: reached (Chain 8), not excluded.** That Kerr admits no conformally-flat slicing [11] bounds only the IWM/CFC *reconstruction*, not the ontology: Kerr does admit a maximal ($K = 0$) slice — DTF's proper now — and sourced forward on it, DTF reproduces Kerr's frame-dragging natively (with $N_{\text{dof}} = 2$) and its full $O(a^2)$ geometry coefficient-exactly (Chain 8), the tensor operator being fixed by Lorentz invariance rather than assumed. And it closes: because the physical frame field is a two-polarisation spin-2 metric ($SO(3)$ internal gauge, $\pi_u \equiv 0$), Deser's uniqueness forces every vertex to GR's, so $\text{DTF} = \text{Kerr}$ to *all orders* in the exterior (computed forward at $O(a^2)$ and $O(a^3)$; the rest inherited from Deser, not computed order-by-order). What remains open is only the *interior* of a rotating hole (rotating collapse / de-emergence), which Chain 8 does not claim.
- **Owed.** The $O(1)$ core coefficient; an anisotropic (BKL) illustration; the full nonlinear closure of $S_u + S_m + S_{\text{frame}}$. (*Not owed*: the magnitude of inertia — mass *is* the ticking rate, so there is no self-energy to compute; only the dimensionless rate *ratios*, the hierarchy, remain, and those every theory owes alike. Chain 4.)

Volume I builds the ontology; **Volume III** reads the quantum. The three volumes are one *organizing principle* (Volume I) checked in three sectors — and this volume's finite core is not a separate result but the *same* clock-definiteness surface as quantum measurement (Volume III), crossed in the opposite direction: there a clock *becomes* definite, here it *de-emerges*. The singularity and the measurement event are one transition, run both ways. A fourth paper — deep cosmology (the proper now as cosmic time) — is in preparation.

Provenance of the checks (ontology-first). This volume carries the trilogy’s genuinely **DTF-native** computational content: the de-emergence surface and fat-core scaling (DTF_singularity_deemergence), the profile-independent $0.23t_P$ collapse halt (DTF_LTB_collapse), and the remnant / no-echo observables (DTF_singularity_observable) — real results, with Hawking evaporation [1] invoked only as an external Row-2 input. The rotating sector (Chain 8) is likewise **computed forward** — though *confirmatory rather than discriminating*, since the Deser closure already fixes the vertices, so these agreements could not have come out otherwise; their value is as a self-test of the construction, and it passes (to $10^{-126}/10^{-131}$): the frame-dragging coefficient (DTF_kerr_constructive), the order-by-order Cotton census on the maximal slice (DTF_kerr_forward_tensor, self-tested to give zero at $O(a^0)$ and $O(a^1)$), the coefficient-exact $O(a^2)$ and $O(a^3)$ balances $\delta R[g_n] = -\delta^{\geq 2} R[g_{<n}]$ (DTF_kerr_oa2_sufficiency, DTF_kerr_oa3_balance), the one-operator identity $\delta R[h^{\text{TT}}] = -\frac{1}{2}\square h$ (DTF_kerr_premise_oneoperator), and the Deser closure (DTF_kerr_deser_closure) — Kerr entering only as the target compared against, never as an input to the derivation. The provenance is graded honestly: the rotating sector is computed forward *through* $O(a^3)$ (confirmatory, per above); the extension to all orders is *inherited* from the Deser bootstrap (shared by any two-polarisation spin-2 theory), a theorem, not an order-by-order DTF computation. The gravity chains (1–4) are **consistency**: GR is *proposed as* what the clock field is made of (conformally-flat-approximate, g_{rr} open), and the two-polarisation count is standard linearised gravity. Nothing here rests on circular reasoning.

Appendix A — The DTF Core in brief

For readers coming to this volume first. The Distorted Time Framework takes the local *rate* of time as the one physical primitive, and builds outward in three layers (Volume I).

- **Row 0 — the timeless substrate.** $\hat{H}\Psi = 0$, no external time; correlations simultaneous, not ordered.
- **Row 1 — the clock.** A definite clock-rate field $u(\mathbf{x}) > 0$; its reading $\tau = \int u dt$ accumulates monotonically — time, and its arrow. Gravity, inertia, and phase live here.
- **Row 2 — space.** An extent is a period radiated at c ($\lambda_C = cT_C$): space is time’s radiation, compiled by a frame field. Because c is one universal exchange rate, Lorentz invariance is its signature.
- **One object, four faces.** Extremising the interval (Fermat, applied to time) gives one phase read four ways: inertia ($\mathbf{p} = \nabla S$), free fall (geodesic), the pilot wave ($\mathbf{v} = \nabla S/m$), and the path integral. Inertia (the rate ω_C) and the pilot wave (the space-face gradient ∇S) are tied by $\mathbf{p} = \nabla S = m\mathbf{v}$, both $\hbar$ -set; the G -coupled well the rate digs is gravity, a separate object.
- **Non-propagating time.** u enters the action with no time-derivative, so $\pi_u \equiv 0$: zero propagating degrees of freedom, and a preferred slice (the *proper now*) that is real but empirically inert. Space radiates (two tensor polarisations, read forward from the extent structure); time does not (no breathing mode).
- **Definiteness.** A single clock-definiteness threshold governs the black-hole core (this volume) and quantum measurement (Volume III).

Code and data availability

The SymPy/NumPy verification scripts named throughout (DTF_singularity_deemergence, DTF_LTB_collapse, DTF_singularity_observable, and the six-script Kerr suite named in the Provenance box — DTF_kerr_constructive, DTF_kerr_forward_tensor, DTF_kerr_oa2_sufficiency, DTF_kerr_oa3_balance, DTF_kerr_premise_oneoperator, DTF_kerr_deser_closure) are publicly available at <https://github.com/billd-dtf/Distorted-Time-Framework>.

Author’s Note: the role of AI

The Distorted Time Framework—its central hypothesis (time’s local *rate*, the ADM lapse, as the one physical primitive), its ontology-first strategy, and the physical interpretations—was conceived and developed by the author over several years of independent research outside academic institutions. The mathematical formalization was carried out in collaboration with Claude (Anthropic): the gravitational field equations and degree-of-freedom count, the black-hole singularity de-emergence and Planck-remnant calculations, the rotating-sector (Kerr) derivations, and the symbolic/numerical verification scripts (SymPy/NumPy) referenced throughout. Separately, other large language models (ChatGPT, Grok, Gemini) were used as independent referees, providing adversarial review of the manuscript; their critiques shaped the revisions but not the derivations.

In each case the author specified the physical target, the constraints the result had to satisfy, and the direction of the argument, and reviewed and approved each result; the AI executed and checked the formal derivations. Without that assistance the author could not have produced the mathematical formalization at this level of detail.

This is disclosed because intellectual honesty requires it. It does not change the epistemic status of the results—a derivation is correct or it is not, independent of how it was produced—but it does mean they warrant careful independent scrutiny rather than assumed authority. The author welcomes that scrutiny; the explicit claim-type labels and the reproducible verification scripts are provided precisely to enable it.

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